PCA and LDA

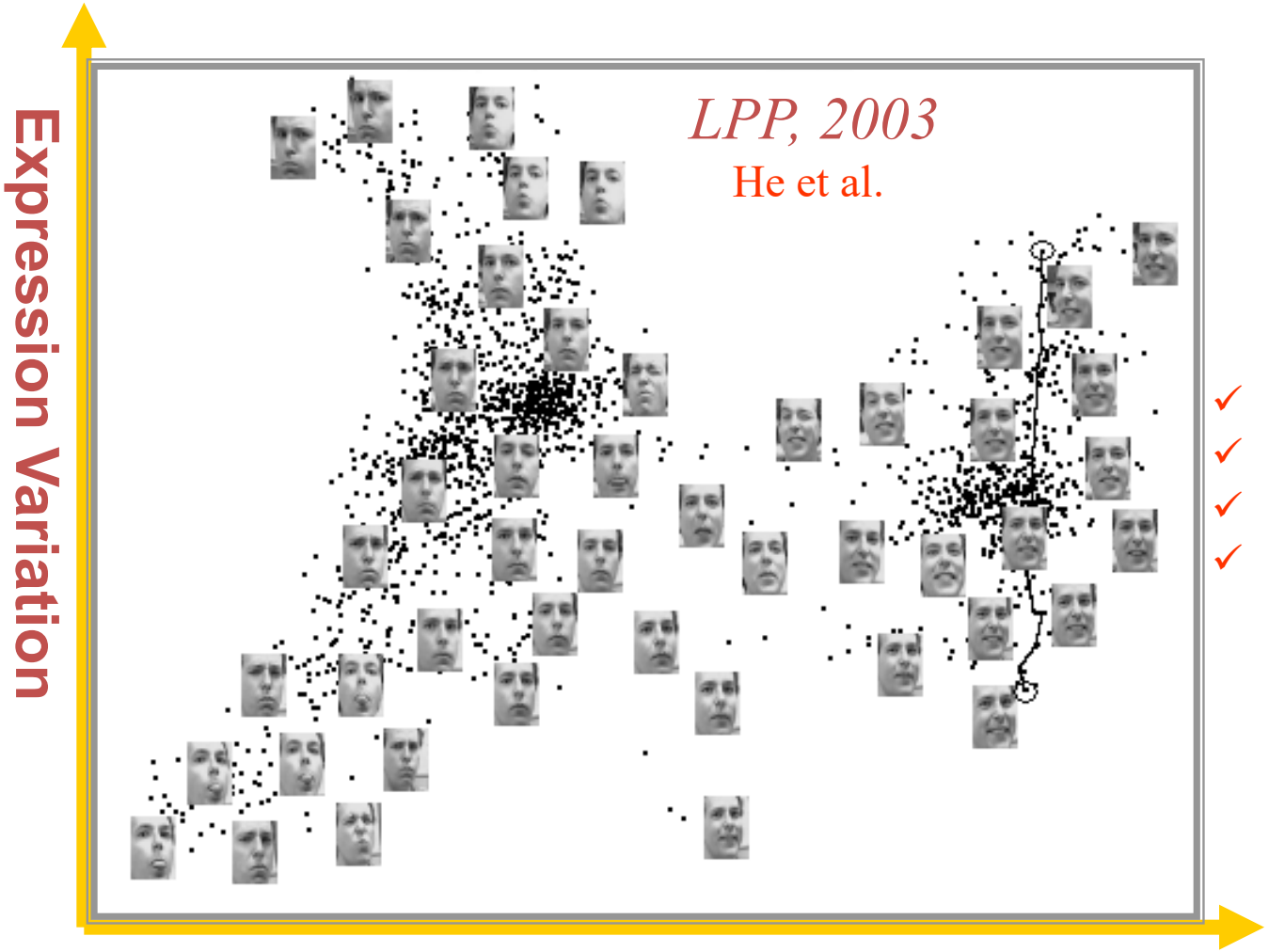
COMP7404
Computational Intelligence and Machine Learning

Dong Xu

The space of all face images

- When viewed as vectors of pixel values, face images are extremely high-dimensional
  - 100x100 image = 10,000 dimensions
  - Slow and lots of storage
- But very few 10,000-dimensional vectors are valid face images
- We want to effectively model the subspace of face images

Why Conduct Dimensionality Reduction?



- ✓ **Visualization**
- ✓ **Feature Extraction**
- ✓ **Computation Efficiency**
- ✓ **Broad Applications**

Face Recognition
Human Gait Recognition
CBIR

Uncover intrinsic structure

Dimension reduction

- Two approaches are available to perform dimensionality reduction
  - **Feature extraction**: creating a subset of new features by combinations of the existing features
  - **Feature selection**: choosing a subset of all the features (the ones more informative)

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \xrightarrow{\text{feature selection}} \begin{bmatrix} x_{i_1} \\ x_{i_2} \\ \vdots \\ x_{i_M} \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \xrightarrow{\text{feature extraction}} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{bmatrix} = f\left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \right)$$

- The problem of feature extraction can be stated as
  - Given a feature space $\boldsymbol{x_i} \in R^N$ find a mapping **y=f(x):** $R^N \rightarrow R^M$ with $M<N$ such that the transformed feature vector $\boldsymbol{y_i} \in R^M$ preserves (most of) the information or structure in R^N.
  - An **optimal** mapping ***y*=f(*x*)** will be one that results in **no increase in the minimum probability of error**
    - This is, a Bayes decision rule applied to the initial space R^N and to the reduced space R^M yield the same classification rate

Dimension reduction

- In general, the optimal mapping y=f(x) will be a non-linear function
  - However, there is no systematic way to generate non-linear transforms
    - The selection of a particular subset of transforms is problem dependent
  - For this reason, feature extraction is commonly limited to linear transforms: ***y=Wx***
    - This is, y is a linear projection of x
    - NOTE: When the mapping is a non-linear function, the reduced space is called a **manifold**

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \xrightarrow{\text{linear feature extraction}} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1N} \\ w_{21} & w_{21} & \cdots & w_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ w_{M1} & w_{M2} & \cdots & w_{MN} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

- We will focus on linear feature extraction.

Principal Component Analysis (PCA)

- Given: N data points $\mathbf{x}_1, \ldots, \mathbf{x}_N$ in R^d
- We want to find a new set of features that are linear combinations of original ones:

$$u(\mathbf{x}_i) = \mathbf{u}^T(\mathbf{x}_i - \boldsymbol{\mu})$$

($\boldsymbol{\mu}$: mean of data points)

- Choose unit vector $\mathbf{u}$ in R^d that captures the most data variance

Principal Component Analysis (PCA)

- Direction that maximizes the variance of the projected data:

Maximize $\frac{1}{N}\sum_{i=1}^{N} \underbrace{\mathbf{u}^{\mathrm{T}}(\mathbf{x}_i - \mu)}(\mathbf{u}^{\mathrm{T}}(\mathbf{x}_i - \mu))^{\mathrm{T}}$ subject to ||**u**||=1

Projection of data point

$$= \mathbf{u}^{\mathrm{T}} \underbrace{\left[1/N \sum_{i=1}^{N} (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^{\mathrm{T}}\right]} \mathbf{u}$$

Covariance matrix of data

$$= \mathbf{u}^{\mathrm{T}} \Sigma \mathbf{u}$$

The direction that maximizes the variance is the eigenvector associated with the largest eigenvalue of Σ

Implementation issue

- Covariance matrix is huge ($d=M^2$)
- But typically # examples << d
- Simple trick
  - **X** is dxN matrix of normalized training data
  - Solve for eigenvectors **u** of $\mathbf{X}^T\mathbf{X}$ instead of $\mathbf{X}\mathbf{X}^T$
  - Then **Xu** is eigenvector of covariance $\mathbf{X}\mathbf{X}^T$
  - Need to normalize each vector of **Xu** into unit length

Eigenfaces (PCA on face images)

1. Compute the principal components ("eigenfaces") of the covariance matrix

$$\boldsymbol{X} = [(\boldsymbol{x_1} - \boldsymbol{\mu}), (\boldsymbol{x_2} - \boldsymbol{\mu}), \ldots \quad , (\boldsymbol{x_N} - \boldsymbol{\mu})]$$
$$[\boldsymbol{U}, \boldsymbol{\lambda}] = \mathrm{eig}(\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X})$$
$$\boldsymbol{V} = \boldsymbol{X}\boldsymbol{U}$$

2. Keep K eigenvectors with largest eigenvalues

$$\boldsymbol{V} = \boldsymbol{V}(:, \mathrm{largest\_eig})$$

3. Represent all face images in the dataset as linear combinations of eigenfaces
   - Perform nearest neighbor on these coefficients

$$\boldsymbol{X_{pca}} = \boldsymbol{V}\left(:, \mathrm{largest_{eig}}\right)^{\boldsymbol{T}}\boldsymbol{X}$$

M. Turk and A. Pentland, Face Recognition using Eigenfaces, CVPR 1991

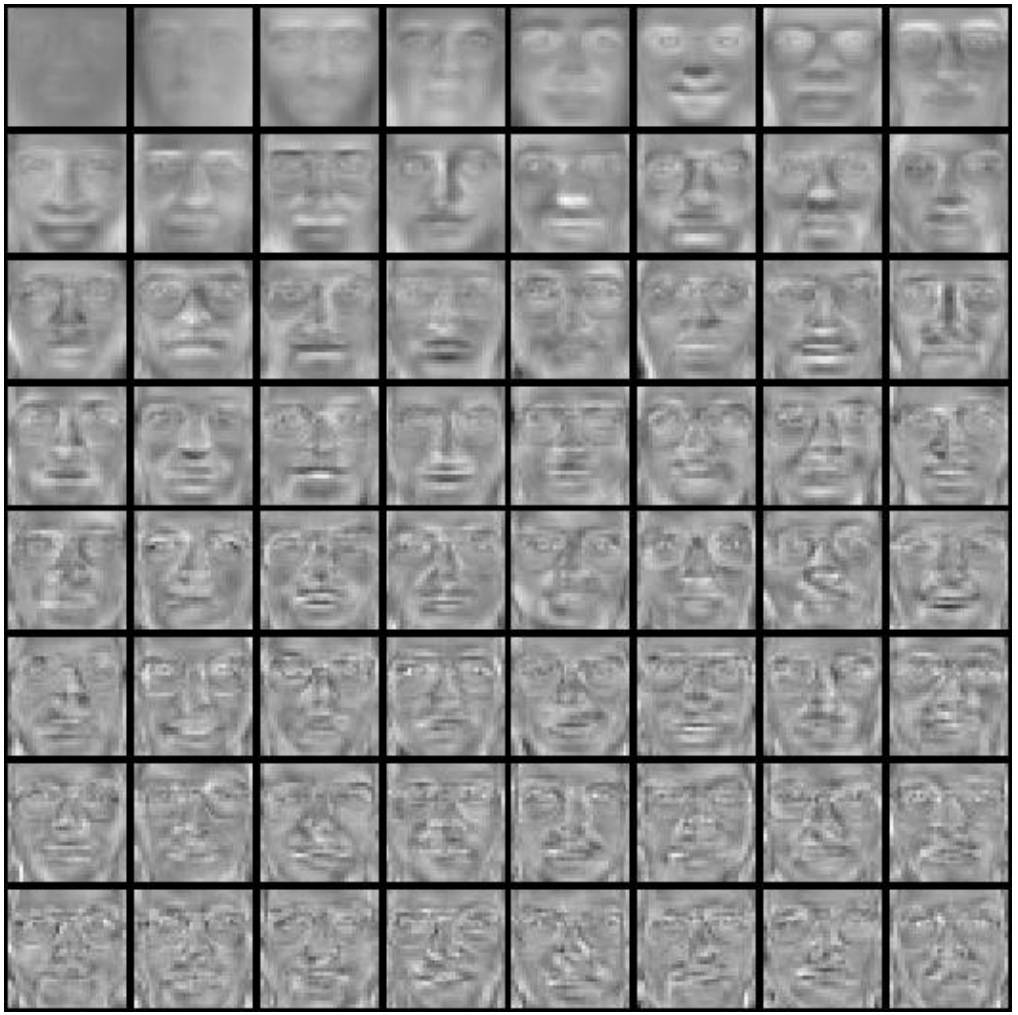
Eigenfaces example

- Training images
- $\mathbf{x}_1,\ldots,\mathbf{x}_N$

Eigenfaces example

Mean: μ

Top eigenvectors: $u_1,\ldots u_k$



Representation and reconstruction

- Face **x** in "face space" coordinates:

$$\mathbf{x} \rightarrow [\mathbf{u}_1^{\mathrm{T}}(\mathbf{x} - \mu), \ldots, \mathbf{u}_k^{\mathrm{T}}(\mathbf{x} - \mu)]$$

$$= \quad w_1, \ldots, w_k$$

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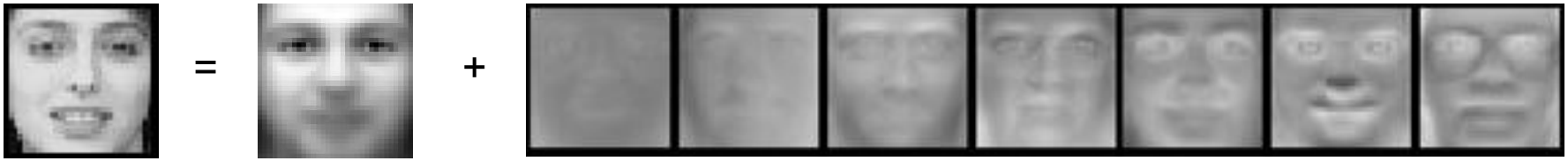
$$= \quad w_1, \ldots, w_k$$

- Reconstruction:



$\hat{x} = \mu + w_1 u_1 + w_2 u_2 + w_3 u_3 + w_4 u_4 + \ldots$

Reconstruction

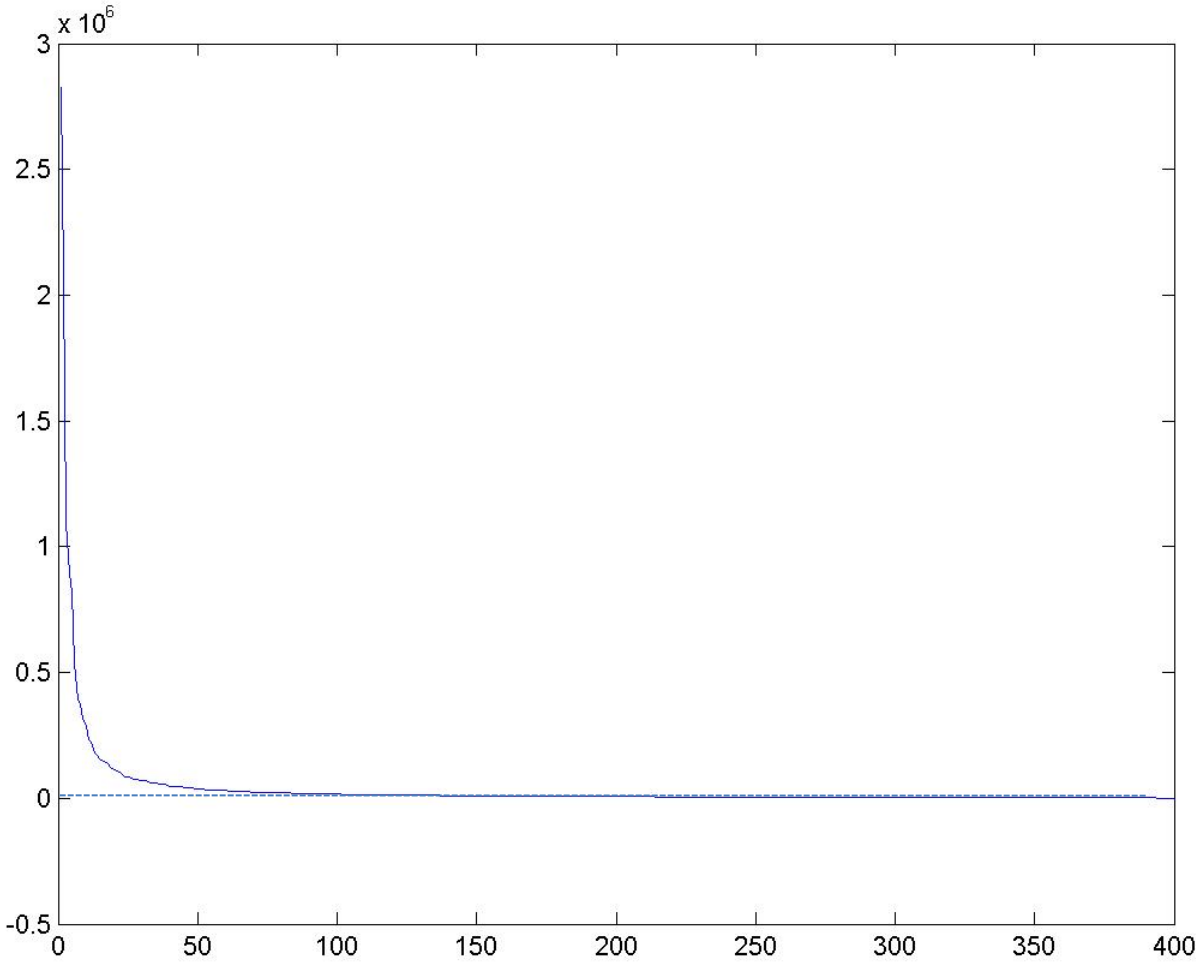
P = 4

P = 200

P = 399

After computing eigenfaces using 399 face images from ORL face database

Eigenvalues (variance along eigenvectors)



PCA: An Example

- Compute the principal components for the following two-dimensional dataset
  - **X=(x1,x2)={(1,2),(3,3),(3,5),(5,4),(5,6),(6,5),(8,7),(9,8)}**
    - Let's first plot the data to get an idea of which solution we should expect
- SOLUTION (by hand)
  - The (biased) covariance estimate of the data is:

$$\Sigma_x = \begin{bmatrix} 6.25 & 4.25 \\ 4.25 & 3.5 \end{bmatrix}$$

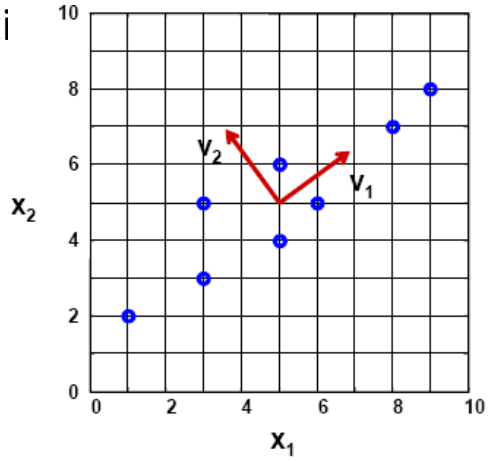
  - The eigenvalues are the zeros of the characteristic equati

$$\Sigma_x v = \lambda v \Rightarrow |\Sigma_x - \lambda I| = 0 \Rightarrow \begin{vmatrix} 6.25-\lambda & 4.25 \\ 4.25 & 3.5-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 9.34;\ \lambda_2 = 0.41;$$

  - The eigenvectors are the solutions of the system

$$\begin{bmatrix} 6.25 & 4.25 \\ 4.25 & 3.5 \end{bmatrix}\begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} \lambda_1 v_{11} \\ \lambda_1 v_{12} \end{bmatrix} \Rightarrow \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0.81 \\ 0.59 \end{bmatrix}$$

$$\begin{bmatrix} 6.25 & 4.25 \\ 4.25 & 3.5 \end{bmatrix}\begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = \begin{bmatrix} \lambda_2 v_{21} \\ \lambda_2 v_{22} \end{bmatrix} \Rightarrow \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = \begin{bmatrix} -0.59 \\ 0.81 \end{bmatrix}$$



Dimension Reduction and Reconstruction

- Let us look at the first data point (1,2)
- Dimension reduction using the projection matrix

$$(0.81 \quad 0.59)\begin{pmatrix}1\\2\end{pmatrix} = 1.99,$$ first dimension using the leading eigenvector

$$(-0.59 \quad 0.81)\begin{pmatrix}1\\2\end{pmatrix} = 1.03,$$ second dimension using the second eigenvector

- Reconstruction
  - Reconstruction using the leading eigenvector only

$$1.99\begin{pmatrix}0.81\\0.59\end{pmatrix} = \begin{pmatrix}1.6119\\1.1741\end{pmatrix}$$

  - Reconstruction using both eigenvectors

$$1.99\begin{pmatrix}0.81\\0.59\end{pmatrix} + 1.03\begin{pmatrix}-0.59\\0.81\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$$

Recognition with eigenfaces

Process labeled training images

- Find mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$
- Find k principal components (eigenvectors of $\boldsymbol{\Sigma}$) $\mathbf{u}_1,\ldots\mathbf{u}_k$
- Project each training image $\mathbf{x}_i$ onto subspace spanned by principal components:
  $(w_{i1},\ldots,w_{ik}) = (\mathbf{u}_1^T(\mathbf{x}_i - \boldsymbol{\mu}), \ldots, \mathbf{u}_k^T(\mathbf{x}_i - \boldsymbol{\mu}))$

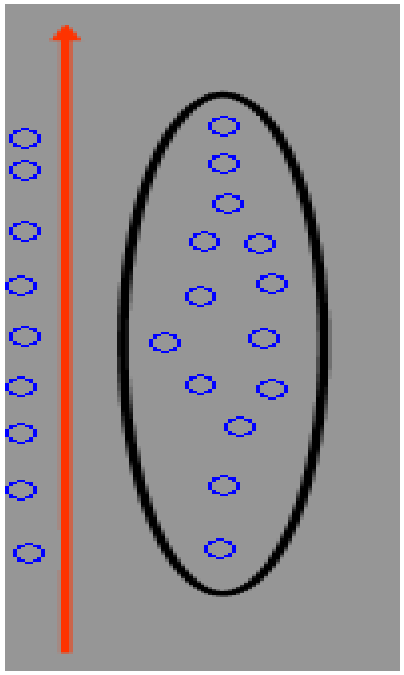
Given novel image $\mathbf{x}$

- Project onto subspace:
  $(w_1,\ldots,w_k) = (\mathbf{u}_1^T(\mathbf{x} - \boldsymbol{\mu}), \ldots, \mathbf{u}_k^T(\mathbf{x} - \boldsymbol{\mu}))$
- Optional: check reconstruction error $\mathbf{x} - \hat{\mathbf{x}}$ to determine whether image is really a face
- Classify as closest training face in k-dimensional subspace

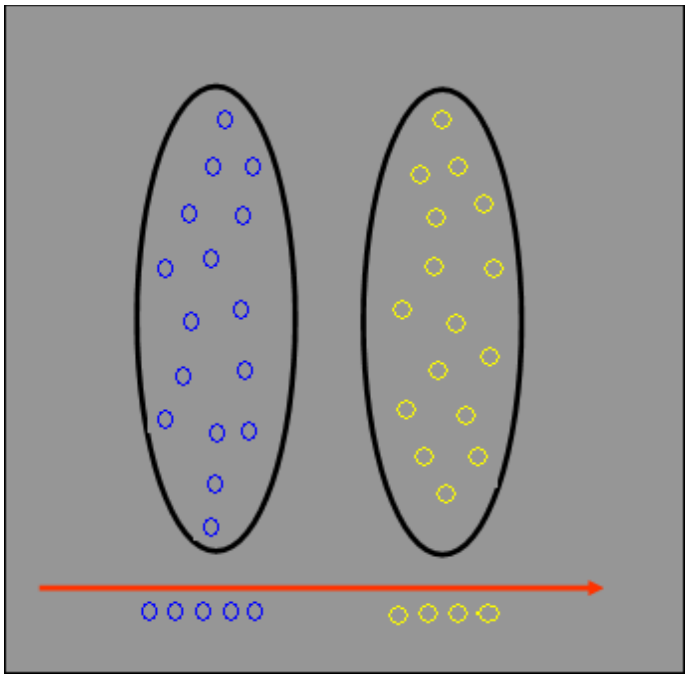
M. Turk and A. Pentland, Face Recognition using Eigenfaces, CVPR 1991

PCA: Limitations

- The direction of maximum variance is not always good for classification



PCA



LDA

Examples: 2D space to 1D space

Principal Components Analysis

- The objective of PCA is to perform dimensionality reduction while preserving as much of the randomness (variance) in the high-dimensional space as possible
  - Let x be an N-dimensional random vector, represented as a linear combination of orthonormal basis vectors $[\phi_1 | \phi_2 | \ldots | \phi_N]$ as

$$x = \sum_{i=1}^{N} y_i \varphi_i \quad \text{where } \varphi_i \mid \varphi_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

  - Suppose we choose to represent x with only M (M<N) of the basis vectors. We can do this by replacing the components $[\mathbf{y_{M+1}}, \ldots \mathbf{y_N}]^T$ with some pre-selected constants b_i

$$x(M) = \sum_{i=1}^{M} y_i \varphi_i + \sum_{i=M+1}^{N} b_i \varphi_i$$

  - The representation error is then

$$\Delta x(M) = x - x(M) = \sum_{i=1}^{N} y_i \varphi_i - \left(\sum_{i=1}^{M} y_i \varphi_i + \sum_{i=M+1}^{N} b_i \varphi_i \right) = \sum_{i=M+1}^{N} (y_i - b_i) \varphi_i$$

  - We can measure this representation error by the mean-squared magnitude of $\Delta x(M)$
  - Our goal is to find the basis vectors φ_i and constants b_i that minimize this mean-square error

$$\bar{\varepsilon}^2(M) = E\left[|\Delta x(M)|^2 \right] = E\left[\sum_{i=M+1}^{N} \sum_{j=M+1}^{N} (y_i - b_i)(y_j - b_j) \varphi_i^T \varphi_j \right] = \sum_{i=M+1}^{N} E\left[(y_i - b_i)^2 \right]$$

  -

Principal Components Analysis

- The optimal values of b_i can be found by computing the partial derivative of the objective function and equating it to zero

$$\frac{\partial}{\partial b_i} E\left[(y_i - b_i)^2\right] = -2\left(E\left[y_i\right] - b_i\right) = 0 \Rightarrow b_i = E\left[y_i\right]$$

  - Therefore, we will replace the discarded dimensions $\mathbf{y_i}$'s by their expected value (an intuitive solution)
- The mean-square error can then be written as

$$\bar{\varepsilon}^2(M) = \sum_{i=M+1}^{N} E\left[\left(y_i - E\left[y_i\right]\right)^2\right] = \sum_{i=M+1}^{N} E\left[\left(x^T\varphi_i - E\left[x^T\varphi_i\right]\right)^T\left(x^T\varphi_i - E\left[x^T\varphi_i\right]\right)\right]$$

$$= \sum_{i=M+1}^{N} \varphi_i^T E\left[(x - E[x])(x - E[x])^T\right]\varphi_i = \sum_{i=M+1}^{N} \varphi_i^T \Sigma_x \varphi_i$$

  - where Σx is the covariance matrix of *x*
- We seek to find the solution that minimizes this expression subject to the orthonormality constraint, which we incorporate into the expression using a set of Lagrange multipliers λ_i

$$\bar{\varepsilon}^2(M) = \sum_{i=M+1}^{N} \varphi_i^T \Sigma_x \varphi_i + \sum_{i=M+1}^{N} \lambda_i(1 - \varphi_i^T\varphi_i)$$

- Computing the partial derivative with respect to the basis vectors

$$\frac{\partial}{\partial \varphi_i}\bar{\varepsilon}^2(M) = \frac{\partial}{\partial \varphi_i}\left[\sum_{i=M+1}^{N} \varphi_i^T \Sigma_x \varphi_i + \sum_{i=M+1}^{N} \lambda_i(1 - \varphi_i^T\varphi_i)\right] = 2\left(\Sigma_x \varphi_i - \lambda_i\varphi_i\right) = 0 \Rightarrow \Sigma_x \varphi_i = \lambda_i\varphi_i$$

$$NOTE: \frac{d}{dx}\left(x^T Ax\right) = \left(A + A^T\right)x \overset{\text{If A is symmetric}}{=} 2\text{Ax}$$

  - So φ_i and λ_i are the eigenvectors and eigenvalues of the covariance matrix Σx

A more discriminative subspace: FLD

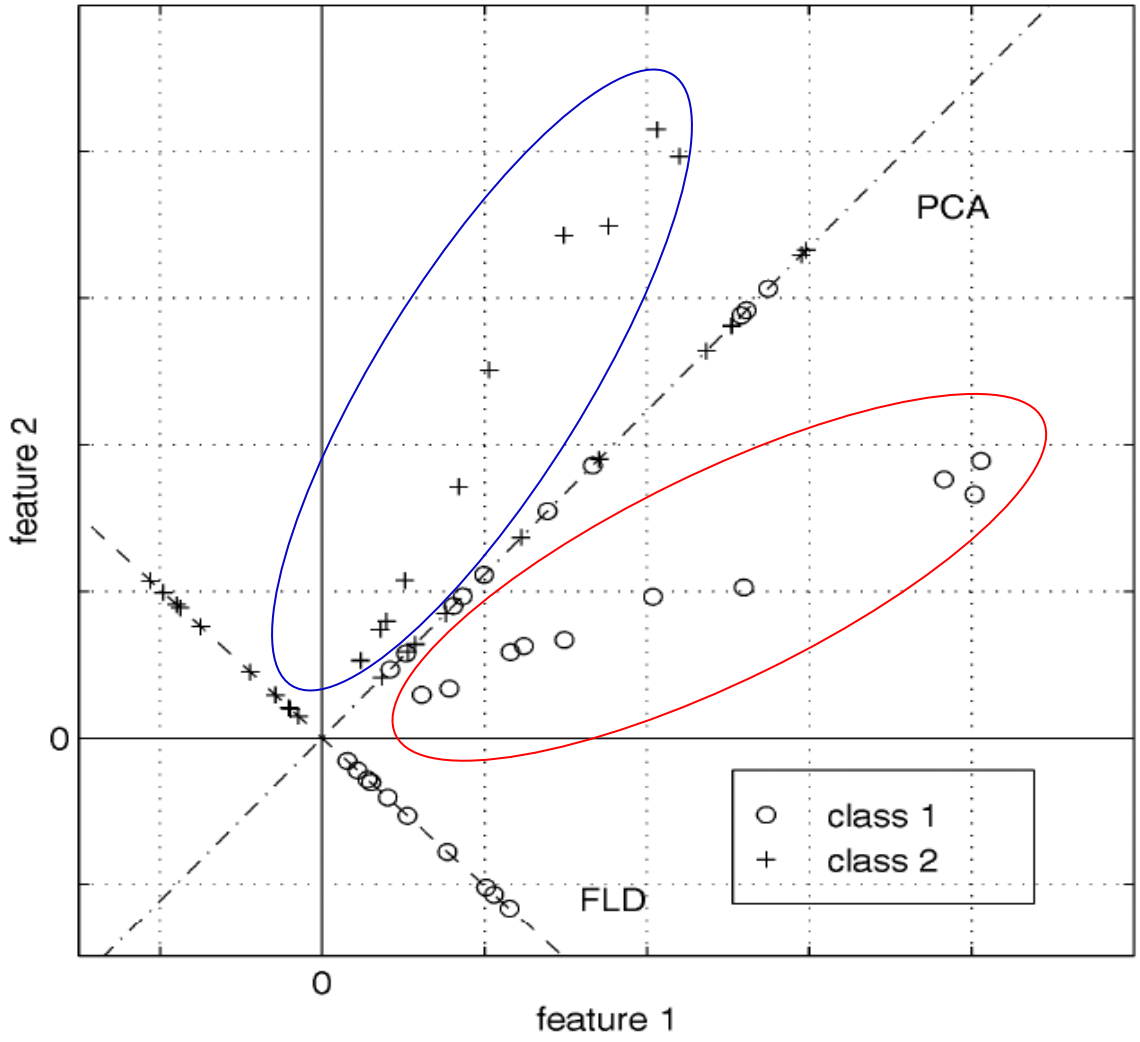
- Fisher Linear Discriminants → "Fisher Faces"

- PCA preserves maximum variance

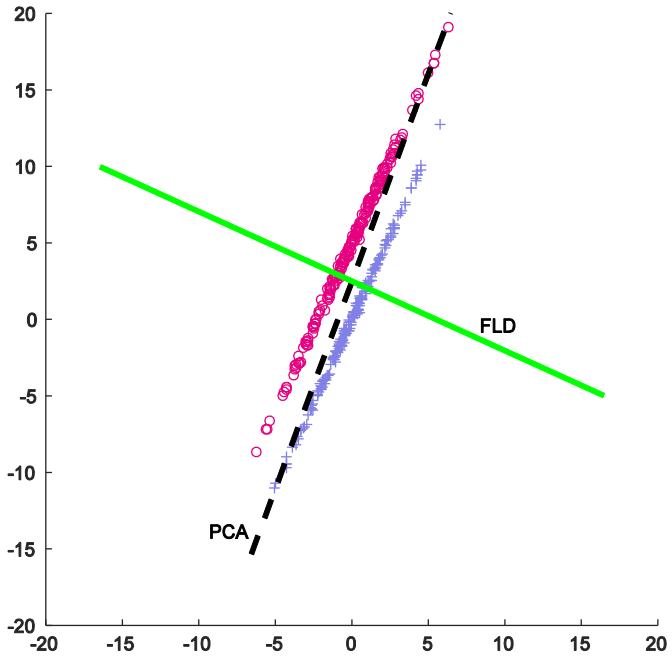
- FLD preserves discrimination
  - Find projection that maximizes scatter between classes and minimizes scatter within classes

Reference: Eigenfaces vs. Fisherfaces, Belheumer et al., PAMI 1997

Comparing with PCA



Idea: motivation of FLD



Bad linear feature (PCA)

Good linear feature (FLD)

Mathematical formulation

- Two categories $y_i \in \{1,2\}$, data $\boldsymbol{x}_i$, N_1, N_2 examples in each category
- Hope to find a projection direction, $u = \boldsymbol{w}^T \boldsymbol{x}$, such that after the projection, examples from two categories are separate

- The means of two categories are
  - $\boldsymbol{\mu}_1 = \frac{1}{N_1} \sum_{y_i=1} \boldsymbol{x}_i$,
  - $\boldsymbol{\mu}_2 = \frac{1}{N_2} \sum_{y_i=2} \boldsymbol{x}_i$
  - The means after projection are $m_1 = \boldsymbol{w}^T \boldsymbol{\mu}_1$, $m_2 = \boldsymbol{w}^T \boldsymbol{\mu}_2$

Objective: Fisher's Criterion

- How to describe "level of separation"?
- Maximize $(m_2 - m_1)^2$?
  - This value can be infinitely large. How?
  - Add a constraint $\boldsymbol{w}^T\boldsymbol{w} = 1$
  - But, a large value does not mean a better direction. How?
- Fisher's criterion
  - While requiring $|m_2 - m_1|$ to be maximized, also asking the two categories to concentrate rather than dispersed after projection:

$$J(\boldsymbol{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}$$

Measuring separation

- For 1D data, the natural measure is variance ($k = 1,2$)

$$s_k^2 = \sum_{y_i=k} (u_i - m_k)^2$$

  - Called the within class scatter
- $s_1^2 + s_2^2$
  - total within-class scatter
- $s_k^2 = \sum_{y_i=k} (u_i - m_k)^2 = \sum_{y_i=k} \left(\boldsymbol{w}^T (\boldsymbol{x}_i - \boldsymbol{\mu}_k)\right)^2 = \boldsymbol{w}^T \sum_{y_i=k} (\boldsymbol{x}_i - \boldsymbol{\mu}_k)(\boldsymbol{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{w}$
- $(m_2 - m_1)^2 = \boldsymbol{w}^T (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T \boldsymbol{w}$

Scatter matrix

- $S_k = \sum_{y_i=k}(\boldsymbol{x}_i - \boldsymbol{\mu}_k)(\boldsymbol{x}_i - \boldsymbol{\mu}_k)^T$
- Within-class scatter matrix

$$S_W = S_1 + S_2$$

- Between-class scatter matrix

$$S_B = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T$$

- Matrix format for Fisher's criterion
  - $\max J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}},\ s.t. \boldsymbol{w}^T \boldsymbol{w} = 1$
  - This optimization problem is called a generalized Rayleigh quotient

Optimization: How to solve it?

- (Simplification/transformation) Exercise: Use the method of Lagrange multipliers to show that the optimal solutions must satisfy

$$S_B \boldsymbol{w} = \lambda S_W \boldsymbol{w}$$

- This is the generalized eigenvalue problem
  - Obtain generalized eigenvalues and eigenvectors of S_B and S_W
- In fact, we do not need to do that
  - $S_B \boldsymbol{w} = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T \boldsymbol{w} \propto (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$
  - $(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) = \lambda S_W \boldsymbol{w}$!

FLD

1. Compute $\boldsymbol{\mu}_2, \boldsymbol{\mu}_1$
2. Compute S_W
3. Compute $\boldsymbol{w} = S_W^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$
4. Normalize:

$$\boldsymbol{w} \leftarrow \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|}$$

But, if not invertible?

- With few data and/or high dimensionality, S_W very likely not invertible
  - generalized inverse matrix
- S_W is real symmetric & at least p.s.d.
  - $S_W = E\Lambda E^{\mathrm{T}},\ \lambda_{ii} \geq 0$
- Moore–Penrose pseudoinverse
  - If $\lambda_{ii} > 0$, define $\lambda_{ii}^{+} = 1/\lambda_{ii}$, otherwise $\lambda_{ii}^{+} = 0$
  - pseudoinverse of Λ: $\Lambda^{+} = \mathrm{diag}(\lambda_{11}^{+}, \lambda_{22}^{+}, \ldots, \lambda_{dd}^{+})$
  - pseudoinverse of S_W

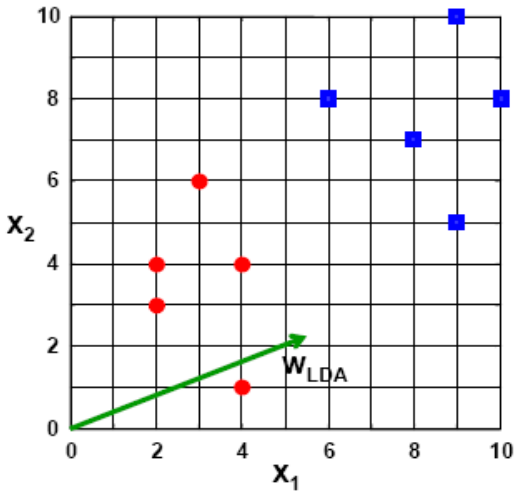
$$S_W^{+} = E\Lambda^{+}E^{T}$$

LDA Example

- Compute the Linear Discriminant projection for the following two-dimensional dataset
  - **X1=(x_1,x_2)={(4,1),(2,4),(2,3),(3,6),(4,4)}**
  - **X2=(x_1,x_2)={(9,10),(6,8),(9,5),(8,7),(10,8)}**
- SOLUTION (by hand)
  - The class statistics are:

$$S_1 = \begin{bmatrix} 0.80 & -0.40 \\ -0.40 & 2.60 \end{bmatrix}; \; S_2 = \begin{bmatrix} 1.84 & -0.04 \\ -0.04 & 2.64 \end{bmatrix}$$

$$\mu_1 = [3.00 \quad 3.60]; \qquad \mu_2 = [8.40 \quad 7.60]$$



  - The within- and between-class scatter are

$$S_B = \begin{bmatrix} 29.16 & 21.60 \\ 21.60 & 16.00 \end{bmatrix}; \; S_W = \begin{bmatrix} 2.64 & -0.44 \\ -0.44 & 5.28 \end{bmatrix}$$

  - The LDA projection is then obtained as the solution of the generalized eigenvalue problem

$$S_W^{-1} S_B v = \lambda v \Rightarrow \left| S_W^{-1} S_B - \lambda I \right| = 0 \Rightarrow \begin{vmatrix} 11.89 - \lambda & 8.81 \\ 5.08 & 3.76 - \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 15.65$$

$$\begin{bmatrix} 11.89 & 8.81 \\ 5.08 & 3.76 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 15.65 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \Rightarrow \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0.91 \\ 0.39 \end{bmatrix}$$

  - Or directly by $w^* = S_W^{-1}(\mu_1 - \mu_2) = [-0.91 \quad -0.39]^T$

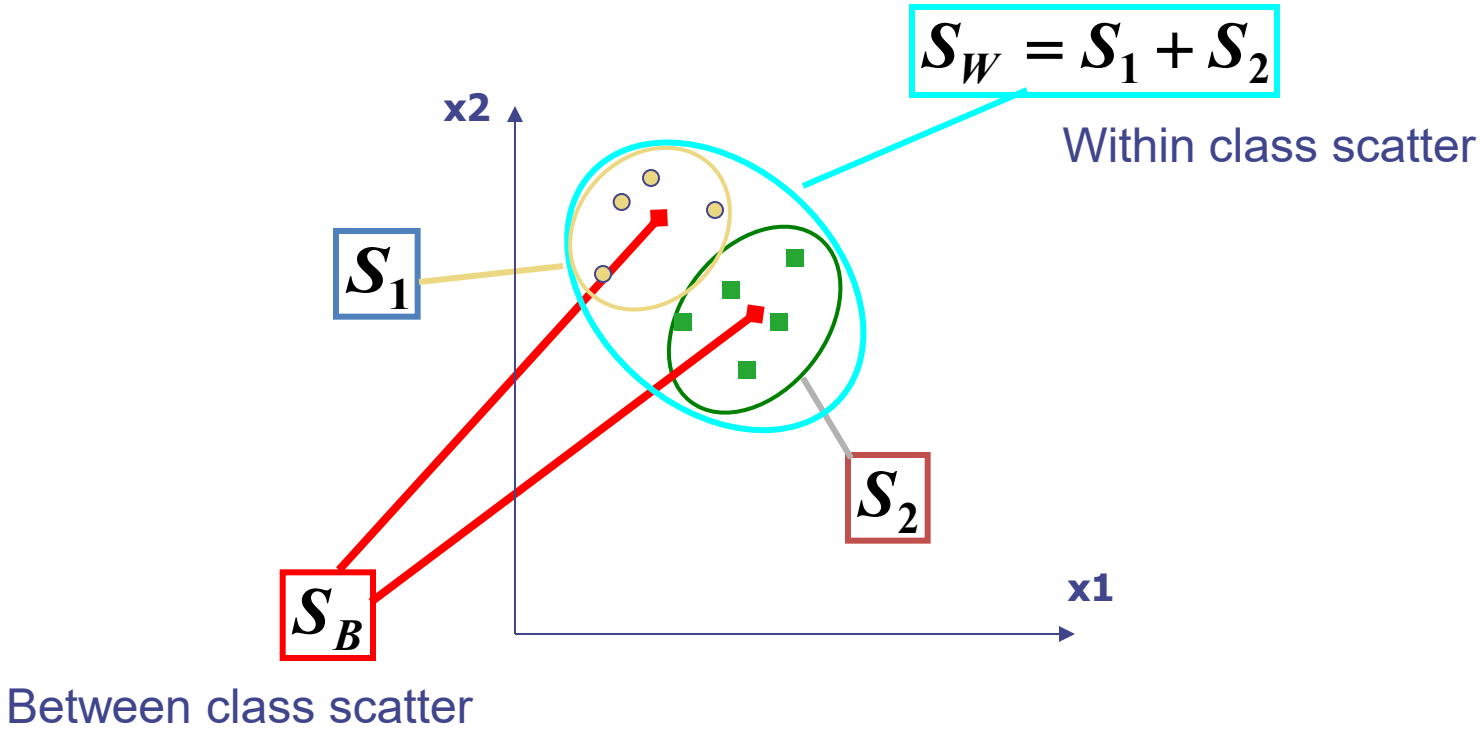
If more than two categories?

- N Sample images: $\{x_1, \cdots, x_N\}$
- c classes: $\{\chi_1, \cdots, \chi_c\}$
- Average of each class: $\mu_i = \frac{1}{N_i} \sum_{x_k \in \chi_i} x_k$
- Average of all data: $\mu = \frac{1}{N} \sum_{k=1}^{N} x_k$

Scatter Matrices

- Scatter of class i: $$S_i = \sum_{x_k \in \chi_i} (x_k - \mu_i)(x_k - \mu_i)^T$$
- Within class scatter: $$S_W = \sum_{i=1}^{c} S_i$$
- Between class scatter: $$S_B = \sum_{i=1}^{c} N_i (\mu_i - \mu)(\mu_i - \mu)^T$$

Illustration



Mathematical Formulation

- After projection $y_k = W^T x_k$
  - Between class scatter $\tilde{S}_B = W^T S_B W$
  - Within class scatter $\tilde{S}_W = W^T S_W W$
- Objective

$$W_{opt} = \arg\max_{W} \frac{|\tilde{S}_B|}{|\tilde{S}_W|} = \arg\max_{W} \frac{|W^T S_B W|}{|W^T S_W W|}$$

- Solution: Generalized Eigenvectors

$$S_B w_i = \lambda_i S_W w_i \qquad i = 1, \ldots, m$$

- Rank of W_{opt} is limited
  - Rank(S_B) <= |C|-1
  - Rank(S_W) <= N-C

Recognition with FLD

- Use PCA to reduce dimensions to N-C

$$W_{pca} = \text{pca}(X)$$

- Compute within-class and between-class scatter matrices for PCA coefficients

$$S_i = \sum_{x_k \in \chi_i} (x_k - \mu_i)(x_k - \mu_i)^T \qquad S_W = \sum_{i=1}^{c} S_i \qquad S_B = \sum_{i=1}^{c} N_i (\mu_i - \mu)(\mu_i - \mu)^T$$

- Solve generalized eigenvector problem

$$W_{fld} = \arg\max_{\text{w}} \frac{\left|W^T S_B W\right|}{\left|W^T S_W W\right|} \qquad S_B w_i = \lambda_i S_W w_i \qquad i = 1, \ldots, m$$

- Project to FLD subspace (c-1 dimensions)

$$W^T{}_{opt} = W^T{}_{fld} W^T{}_{pca} \qquad \hat{x} = W_{opt}{}^T x$$

- Classify by nearest neighbor

Note: x in step 2 refers to PCA coef; x in step 4 refers to original data

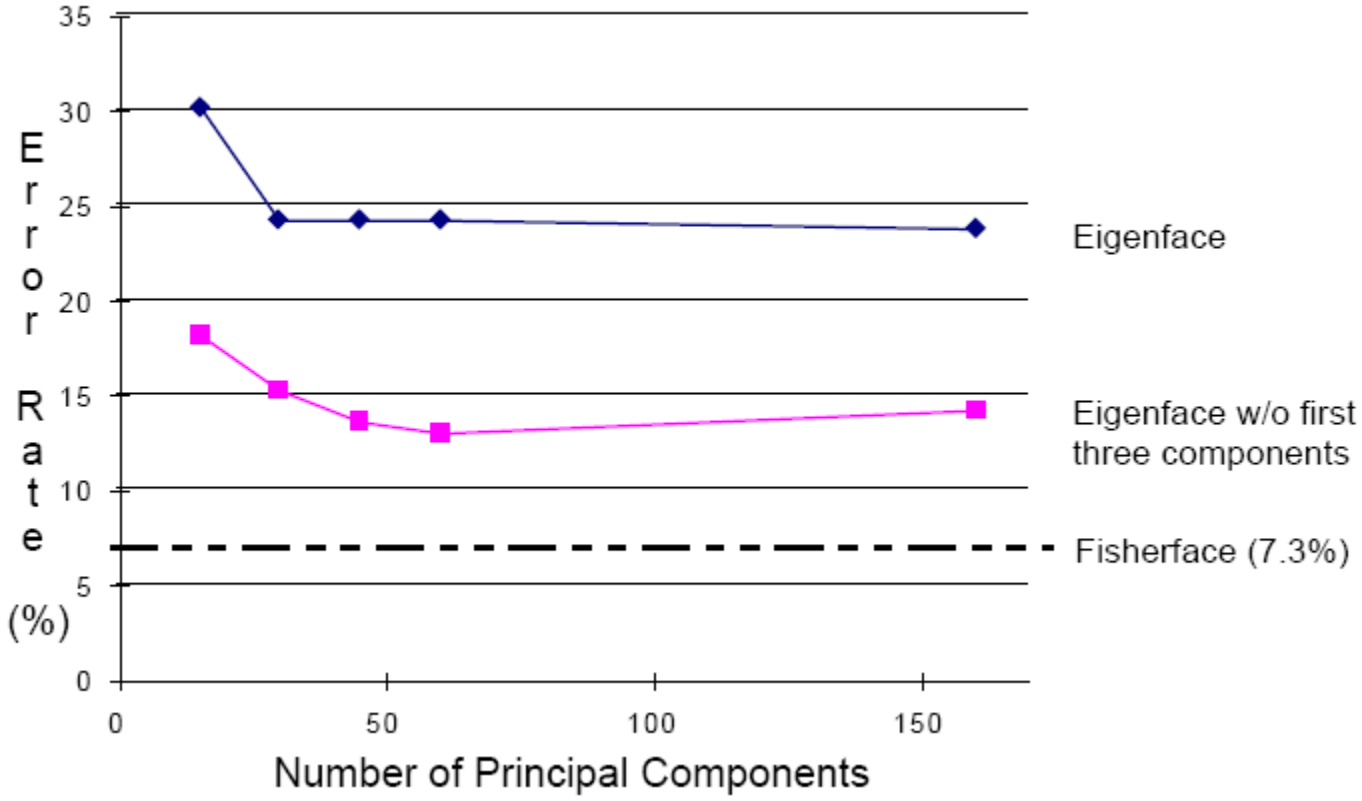
Results: Eigenface vs. Fisherface

- Input: 160 images of 16 people
- Train: 159 images
- Test: 1 image

- Variation in Facial Expression, Eyewear, and Lighting

Reference: Eigenfaces vs. Fisherfaces, Belheumer et al., PAMI 1997

Eigenfaces vs. Fisherfaces



Reference: Eigenfaces vs. Fisherfaces, Belheumer et al., PAMI 1997

Thank you!